

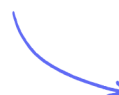
Structured Questions

Work, Energy & Power

Work / Kinetic Energy / Gravitational Potential Energy / The Principle of Conservation of Energy / Power / Efficiency

Medium (1 question)	/9
Hard (1 question)	/9
Total Marks	/18

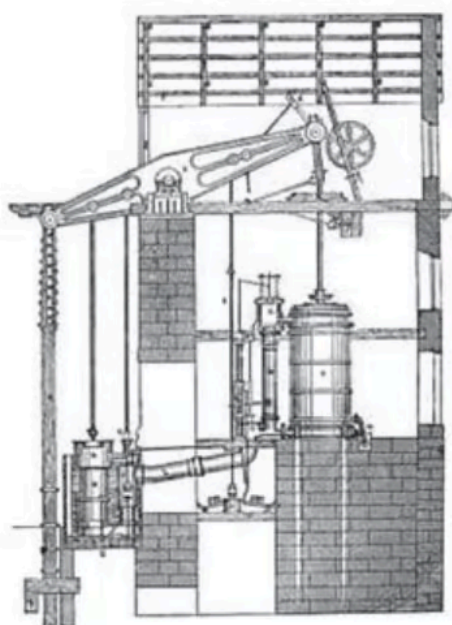
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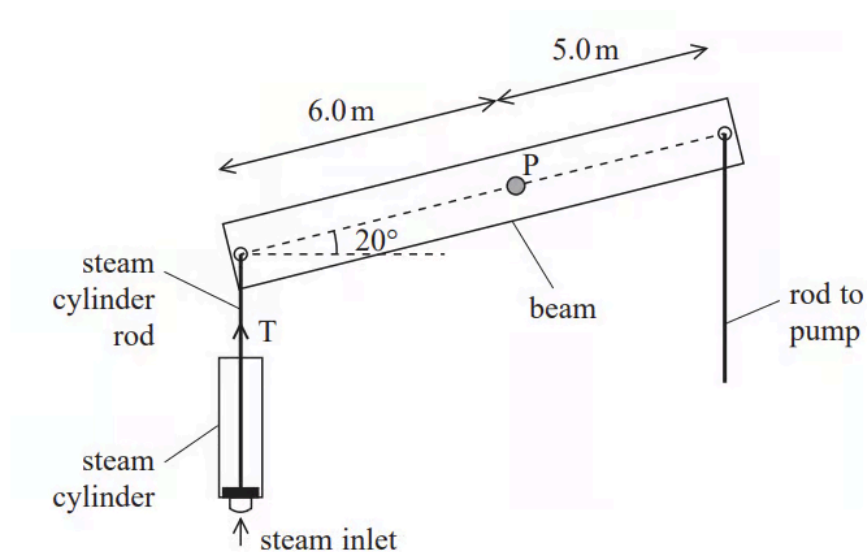
Medium Questions

- 1 (a)** Beam engines contributed to powering the Industrial Revolution in Britain in the 18th century. A beam engine consisted of a beam which could rock to and fro around a well-

oiled pivot. Attached to the beam there are two rods, one connected to a piston in a steam cylinder and the other connected to a pump.



The diagram below shows a simplified arrangement of a beam engine.



The beam has a constant thickness and a mass of 3.05×10^4 kg. The length of the beam is 11.0 m. The pivot P is positioned 6.0 m from the steam cylinder end of the beam.

In its resting position the steam cylinder rod is supported by the base of the steam cylinder with the beam at an angle of 20° to the horizontal.

The steam cylinder rod exerts a force T on the beam. The force exerted on the beam by the pump rod can be neglected.

Calculate the force T .

$T = \dots\dots\dots$

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(4 marks)

- (b)** The engine, which ran continuously, could lift a mass of 2500 kg of water through 12 m each minute.

The engine used 1250 kg of coal a day. 1 kg of coal can release 22.3 MJ of energy.

The beam engine was said to have an efficiency of 10 %.

Deduce whether this claim for efficiency was correct.

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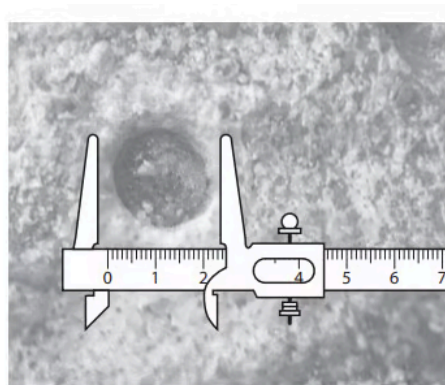
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(5 marks)

Hard Questions

- 1 (a)** Impact craters are formed when meteorites strike the surface of a planet. A student investigated some factors that might influence the formation of impact craters. He did this by dropping spheres of modelling clay into a tray of sand.

The diameter of the crater produced by each sphere was measured using vernier calipers as shown.



This process was repeated for spheres of different diameters.

In one test, the spheres were dropped from the same height.

Determine the factor by which the kinetic energy of the sphere just before impact increases when the sphere diameter is increased from 2.0 cm to 4.0 cm.

Factor =

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(3 marks)

- (b)** The student also dropped the spheres from different heights. His results are shown in the table.

Drop height / m	Sphere diameter / cm	Crater diameter / cm
0.30	2.0	3.6
	4.0	7.0
	6.0	6.8
0.60	2.0	4.8
	4.0	7.5
	6.0	7.3
0.90	2.0	5.6
	4.0	8.0
	6.0	8.3

The student wrote the following conclusion:

“The greater the drop height, the greater the diameter of the crater formed when a sphere hits the sand. This is because the impact velocity increases as the drop height is increased. As the speed of the sphere increases the diameter of the crater formed also increases. Also, the bigger the sphere the bigger the crater.”

Assess the validity of the student’s conclusion.

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(6 marks)